

Clustering Rule Process on iris.arff using Simple K-Means

Specifications Required

- Software: WEKA (Waikato Environment for Knowledge Analysis)
- Dataset: iris.arff (comes pre-installed with WEKA)
- System Requirements: Java Runtime Environment, GUI OS
- Skills: Basic understanding of unsupervised learning and clustering

Aim

To demonstrate the clustering process using the Simple K-Means algorithm on the iris.arff dataset in WEKA.

Theory

Clustering is an unsupervised learning technique that groups similar instances into clusters without prior labels. In WEKA, Simple K-Means is a widely used clustering algorithm.

Key Concepts:

- K-Means Algorithm partitions data into K clusters.
- Each instance is assigned to the cluster with the nearest centroid.
- The centroids are updated iteratively until the algorithm converges.

Steps to Perform in WEKA

1. Open WEKA and go to the Explorer.
2. Click on 'Open file' and load iris.arff.
3. Go to the 'Cluster' tab.
4. Select 'SimpleKMeans' as the clustering algorithm.
5. Click 'Choose' and set the number of clusters to 3.
6. Click 'Start' to run the clustering process.

7. Review the output: cluster sizes, centroids, instance assignments.

Expected Output

- 3 clusters created (matching the 3 iris species).
- Each instance assigned a cluster ID.
- Visualization displays data distribution among clusters.
- Centroids and cluster sizes are shown in the result.

Conclusion

SimpleKMeans successfully clusters the Iris dataset into three groups, closely resembling the actual species. This highlights how unsupervised learning can reveal natural patterns in the data.